

● HARD

● INFORMATION AND IDEAS

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06

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

- A. The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum's long-suspected role in motor coordination.
- B. Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway connecting the VTA and the cerebellum.
- C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.
- D. Technological limitations have historically hindered the study of the cerebellum, but the recent development of new technologies has led to greater insights into its functions.

When digging for clams, their primary food, sea otters damage the roots of eelgrass plants growing on the seafloor. Near Vancouver Island in Canada, the otter population is large and well established, yet the eelgrass meadows are healthier than those found elsewhere off Canada's coast. To explain this, conservation scientist Erin Foster and colleagues compared the Vancouver Island meadows to meadows where otters are absent or were reintroduced only recently. Finding that the Vancouver Island meadows have a more diverse gene pool than the others do, Foster hypothesized that damage to eelgrass roots increases the plant's rate of sexual reproduction; this, in turn, boosts genetic diversity, which benefits the meadows' health overall.

Which finding, if true, would most directly undermine Foster's hypothesis?

- A. At some sites in the study, eelgrass meadows are found near otter populations that are small and have only recently been reintroduced.
- B. At several sites not included in the study, there are large, well-established sea otter populations but no eelgrass meadows.
- C. At several sites not included in the study, eelgrass meadows' health correlates negatively with the length of residence and size of otter populations.
- D. At some sites in the study, the health of plants unrelated to eelgrass correlates negatively with the length of residence and size of otter populations.

Arthurian legends (tales related to the character of King Arthur) derive from many sources, such as *Vita Sancti Cadoci*, composed in the 11th century, and *Culhwch and Olwen* from the second half of the 12th century. One of the most significant sources, Geoffrey of Monmouth's *History of the Kings of Britain*, was written in the 1130s; some material from it was later adapted by the Norman poet Wace into the *Roman de Brut* in 1155. But Wace didn't merely adapt *History*, he added to it as well, introducing the famous Round Table at which Arthur's knights assembled, which suggests that _____ blank

Which choice most logically completes the text?

- A. *Roman de Brut* is more historically accurate than *History*, because *Culhwch and Olwen* had not been written when Geoffrey of Monmouth was writing his work.
- B. Geoffrey of Monmouth was unaware of stories of the Round Table when composing his *History*, though historians know that works containing such stories were available to him.
- C. Geoffrey of Monmouth's accounts of Arthurian legends in his *History* are more similar overall in content to the accounts in *Culhwch and Olwen* than they are to the accounts in *Roman de Brut*.
- D. the Arthurian legends that the author of *Vita Sancti Cadoci* drew on would not have featured the Round Table.

The following text is from Mick Herron’s 2023 novel *The Secret Hours*. The narrator is describing members of a government committee.

Finally, and adding much-needed gravitas, was Sir Winston Day, whose features seemed moulded to adorn a bust, or possibly a stamp, and whose forehead was so evidently bulging with grey matter that it would have been impertinent to inquire too closely into the actual achievements his half century of public service had produced. His recently published memoirs possibly cast light on this enigma, but given that such details were not provided until after the thirty-page mark, they might as well have remained state secrets.

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Based on the text, which choice best describes Sir Winston Day?

- A. He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction.
- B. He looks like a person worthy of respect, but his memoirs reveal that some of his actions were dishonorable.
- C. He would be a celebrated public figure if his achievements did not have to be kept secret.
- D. He has maintained a modest profile even though he has served the public capably for many years.

Biologist Valentina Gómez-Bahamón and her team have investigated two subspecies of the fork-tailed flycatcher bird that live in the same region in Colombia, but one subspecies migrates south for part of the year, and the other doesn't. The researchers found that, due to slight differences in feather shape, the feathers of migratory forked-tailed flycatcher males make a sound during flight that is higher pitched than that made by the feathers of nonmigratory males. The researchers hypothesize that fork-tailed flycatcher females are attracted to the specific sound made by the males of their own subspecies, and that over time the females' preference will drive further genetic and anatomical divergence between the subspecies.

Which finding, if true, would most directly support Gómez-Bahamón and her team's hypothesis?

- A. The feathers located on the wings of the migratory fork-tailed flycatchers have a narrower shape than those of the nonmigratory birds, which allows them to fly long distances.
- B. Over several generations, the sound made by the feathers of migratory male fork-tailed flycatchers grows progressively higher pitched relative to that made by the feathers of nonmigratory males.
- C. Fork-tailed flycatchers communicate different messages to each other depending on whether their feathers create high-pitched or low-pitched sounds.
- D. The breeding habits of the migratory and nonmigratory fork-tailed flycatchers remained generally the same over several generations.

The *Mammillaria* cactus *M. boolii* occurs naturally only in the state of Sonora in Mexico, and the smallness of its range makes it especially vulnerable to extinction. The traditional single-species approach to conservation emphasizes the need to focus on individual species most at risk, like *M. boolii*, but recently, conservationists have argued that an ecosystem-based approach that incorporates the many interactions between the climate, terrain, and various species of a given geographical area may lead to better outcomes for all the species in a given location. If this view is correct, the single-species approach to the conservation of *M. boolii* could thus _____ blank

Which choice most logically completes the text?

- A. lead to a better understanding of how the distribution of *Mammillaria* species throughout Mexico has affected their survival.
- B. allow conservationists to better consider how climatic changes affecting Sonora may reduce the number of species competing with *M. boolii*.
- C. erroneously shift the focus of conservation efforts away from *M. boolii* itself.
- D. fail to consider the ways in which the survival of *M. boolii* may be influenced by changes in the populations of other species that inhabit Sonora.

In 2018, scientists discovered an immense aggregation of *Muusoctopus robustus* (pearl octopuses) along a hydrothermal vent 3,200 meters beneath the ocean's surface. Water temperatures at this site—named the Octopus Garden—climb as high as 11°C, much warmer than the ambient 1.6°C typical at this depth. Based on observations made over three years, scientists concluded that temperatures at the site likely confer reproductive benefits and that the site is used exclusively for reproduction—6,000 *M. robustus* adults, hatchlings, and eggs were observed at the garden, but no juveniles were present.

Which statement about *M. robustus* and the Octopus Garden is best supported by the text?

- A. *M. robustus* leave the Octopus Garden upon reaching an intermediary stage of development.
- B. The *M. robustus* population at the Octopus Garden remains stable despite variations in water temperature.
- C. *M. robustus* nests in the Octopus Garden contain on average fewer but larger eggs than nests at similar ocean depths.
- D. The Octopus Garden provides an ideal feeding ground for *M. robustus* hatchlings.

Employment by Sector in France and the United States, 1800–2012 (%)

Year	Agriculture in France	Manufacturing in France	Services in France	Agriculture in US	
1800	64	22	14	68	
1900	43	29	28	41	
1950	32	33	35	14	
2012	3	21	76	2	

Rows in table may not add up to 100 due to rounding.

Over the past two hundred years, the percentage of the population employed in the agricultural sector has declined in both France and the United States, while employment in the service sector (which includes jobs in retail, consulting, real estate, etc.) has risen. However, this transition happened at very different rates in the two countries. This can be seen most clearly by comparing the employment by sector in both countries in _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. 1900 with the employment by sector in 1950.
- B. 1800 with the employment by sector in 2012.
- C. 1900 with the employment by sector in 2012.
- D. 1800 with the employment by sector in 1900.

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht’s influence can be seen in English playwright Caryl Churchill’s 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____ blank

Which choice most logically completes the text?

- A. focus on the characters’ beliefs about social and political issues as revealed by the characters’ actions.
- B. reflect on social and political phenomena not directly related to patriarchy and colonialism.
- C. recognize pertinent social and political parallels between Germany during Brecht’s time and England at the time when Churchill was writing *Cloud 9*.
- D. be dispassionate as they think critically about the social and political questions raised by the play.

Many literary theorists distinguish between *fabula*, a narrative's content, and *syuzhet*, a narrative's arrangement and presentation of events. In the film *The Godfather Part II*, the *fabula* is the story of the Corleone family, and the *syuzhet* is the presentation of the story as it alternates between two timelines in 1901 and 1958. But literary theorist Mikhail Bakhtin maintained that *fabula* and *syuzhet* are insufficient to completely describe a narrative—he held that systematic categorizations of artistic phenomena discount the subtle way in which meaning is created by interactions between the artist, the work, and the audience.

Which choice best states the main idea of the text?

- A. Literary theorist Mikhail Bakhtin argued that there are important characteristics of narratives that are not fully encompassed by two concepts that other theorists have used to analyze narratives.
- B. Literary theorist Mikhail Bakhtin claimed that meaning is not inherent in a narrative but is created when an audience encounters a narrative so that narratives are interpreted differently by different people.
- C. The storytelling methods used in *The Godfather Part II* may seem unusually complicated, but they can be easily understood when two concepts from literary theory are utilized.
- D. Narratives that are told out of chronological order are more difficult for audiences to understand than are narratives presented chronologically.